



BLOCKCHAIN DEVELOPER ROADMAP

Step-by-Step Guide to Become a Blockchain Developer



LEARN



PRACTICE



BUILD



DEPLOY



GROW

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1. Blockchain Fundamentals

- What is Blockchain?
- How Blockchain Works?
- Types of Blockchains (Public, Private, Consortium)
- Key Features, Benefits & Limitations

1–2 Weeks

Resources

Coursera, Binance Academy, 101 Blockchains

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2. Programming Basics

- Choose a Language: Solidity (for Smart Contracts) OR Go, Rust, Python, JavaScript (Node.js)
- Data Structures, Algorithms (Basics)
- OOP & Problem Solving

2–3 Weeks

Resources

Solidity Docs, MDN, freeCodeCamp

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3. Cryptography Essentials

- Hash Functions
- Public Key & Private Key
- Digital Signatures
- Merkle Trees, Encryption Basics

1–2 Weeks

Resources

Coursera, Khan Academy, CyberChef

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4. How Blockchain Works (Deep Dive)

- Blocks, Transactions, Merkle Root
- Consensus Mechanisms (PoW, PoS, DPoS, etc.)
- Mining, Validation & Nodes
- Gas, Fees & Transaction Lifecycle

2–3 Weeks

Resources

Ethereum.org, Whitepapers, Visual Blockchain

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5. Smart Contracts Development

- What is a Smart Contract?
- Solidity Basics (Variables, Functions, Modifiers)
- Advanced Concepts (Inheritance, Libraries, Events)
- Testing Smart Contracts

3–4 Weeks

Resources

Solidity Docs, CryptoZombies, Dapp University

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6. Blockchain Platforms

- Ethereum Overview
- Comparing Chains: Ethereum, BSC, Polygon, Solana, Avalanche, Hyperledger
- When to use which Blockchain?

2–3 Weeks

Resources

Docs of each platform, YouTube, Blogs

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7. DApp (Decentralized App) Development

- Frontend: HTML, CSS, JavaScript, Web3.js / Ethers.js
- Connect Wallets (MetaMask)
- Read/Write Smart Contracts
- Build Full DApp

3–4 Weeks

Resources

Web3.js Docs, Ethers.js, Dapp University

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8. Storage & Oracles

- On-chain vs Off-chain Storage
- IPFS (InterPlanetary File System)
- Oracles: Chainlink Basics
- Decentralized Storage Solutions

2 Weeks

Resources

IPFS Docs, Chainlink Docs, Filecoin Docs

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9. Security Best Practices

- Smart Contract Security Tips
- Common Vulnerabilities (Reentrancy, Overflow, etc.)
- Auditing Basics
- Tools: Slither, MythX, Remix Analyzer

2 Weeks

Resources

Consensys Diligence, OWASP, Trail of Bits

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10. Testing, Deployment & DevOps

- Testing Frameworks (Hardhat, Truffle, Foundry)
- Deploy Smart Contracts
- CI/CD for Blockchain (GitHub Actions)
- Monitor with Etherscan / Tenderly

2–3 Weeks

Resources

Hardhat Docs, Foundry, Tenderly Docs

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11. Advanced Topics

- Layer 2 Solutions (Optimism, Arbitrum, zkSync)
- Tokens (ERC-20, ERC-721, ERC-1155) & Standards
- DeFi Basics, DAOs, Governance
- Zero Knowledge Proofs (Basics)

3–4 Weeks

Resources

Blogs, Research Papers, Bankless, DeFi Llama

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12. Build Real-World Projects

- Start with Simple Projects
- Build & Deploy Multiple DApps
- Contribute to Open Source
- Build Portfolio & Share Your Work

Ongoing

Resources

GitHub, Awesome Blockchain, Web3 Projects

ESSENTIAL SKILLS

- Blockchain Architecture
- Smart Contract Development
- Distributed Systems
- Cryptography
- Web3 & DApp Development
- Problem Solving & Debugging
- Security Mindset
- Continuous Learning

IMPORTANT TOOLS

- | | | |
|-----------|----------|-----------|
| Remix IDE | MetaMask | Hardhat |
| Truffle | Ganache | Ethers.js |
| Web3.js | IPFS | Solidity |
| Chainlink | Infura | Tenderly |

PROJECT IDEAS

- ✓ Decentralized To-Do App
- ✓ Token (ERC-20) with Staking
- ✓ NFT Marketplace
- ✓ Voting DApp
- ✓ DeFi Lending/Borrowing Platform
- ✓ Crowdfunding DApp
- ✓ DAO with Governance
- ✓ Decentralized Chat App

LEARNING PATH FLOW



CAREER OPPORTUNITIES

- Blockchain Developer
- Smart Contract Developer
- DApp Developer
- DeFi Developer
- Blockchain Solutions Architect
- Web3 Developer
- Blockchain Security Engineer
- Research Analyst



Consistency + Practice + Real Projects = Blockchain Developer Success!

